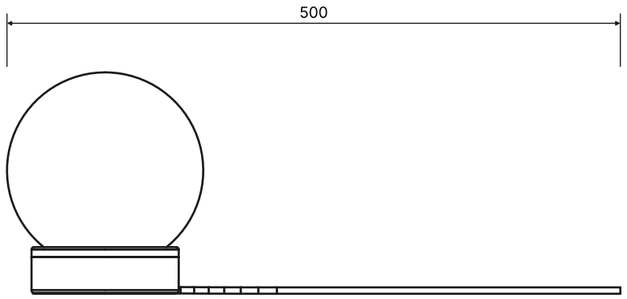
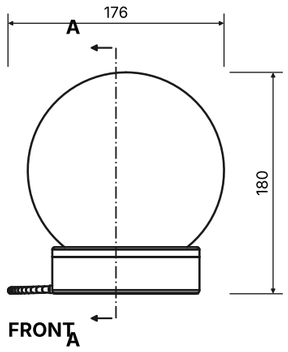
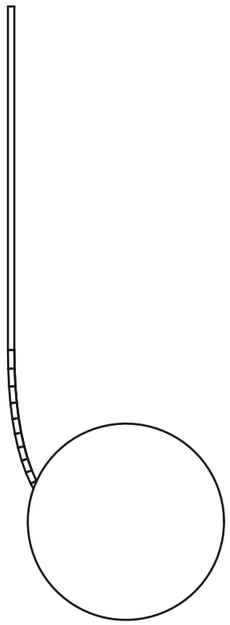
Scale 1:5. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

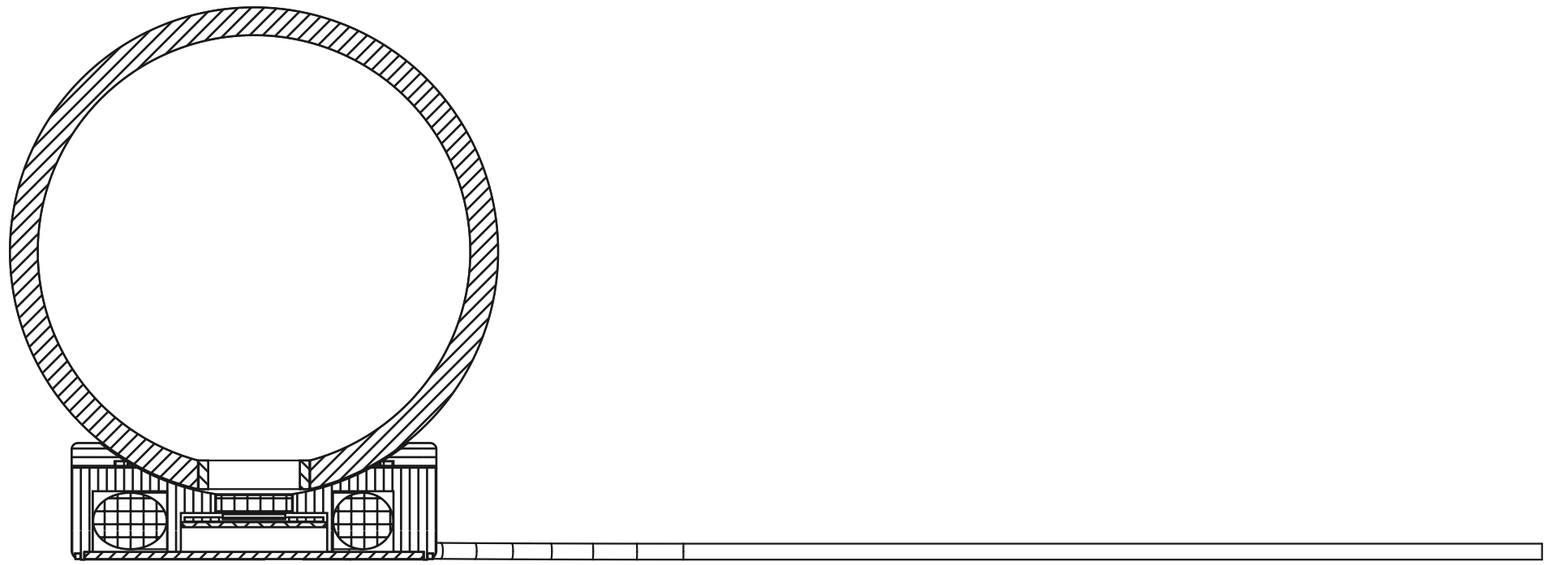
General arrangement

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. Section A-A (sheet 1): the product cut at x -8.1 and seen from the right; cut faces hatched, a different angle for each part.



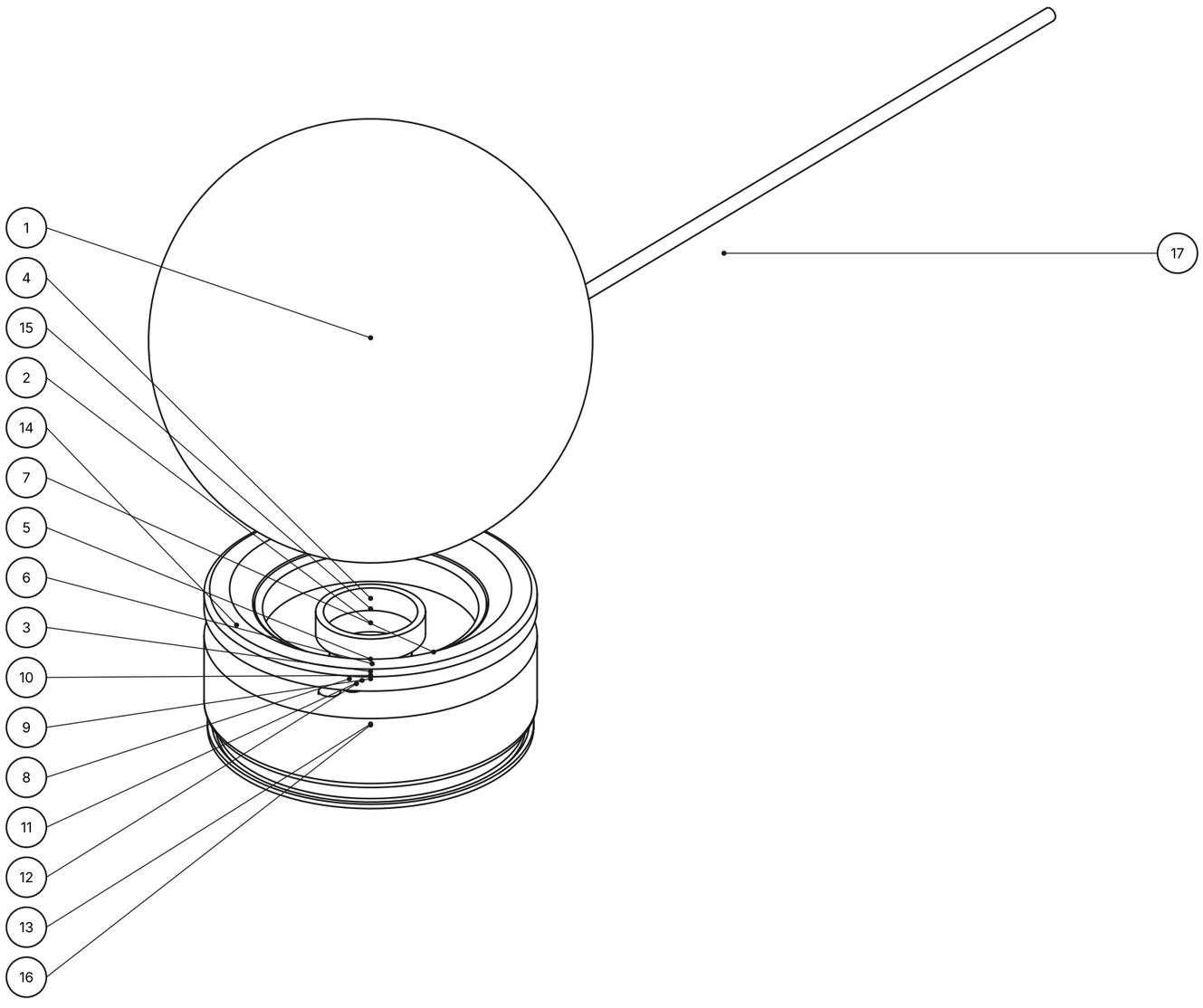
Section A-A

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

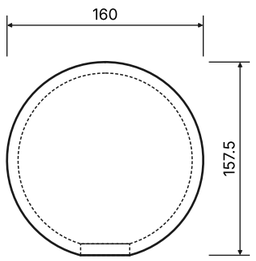
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

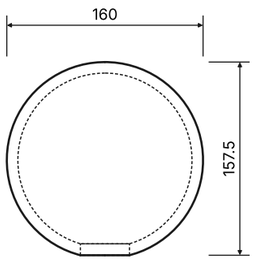
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-formed; 634 cm3 each; 1 for the set; 1:5

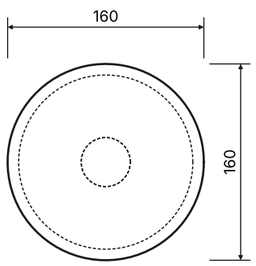
the ball: two hemispheres drape-formed from 10 mm tinted cast sheet on one aluminium mould (the production process; the prototypes are formed on it), their insides bead-blasted, joined at the equator with a polymerising acrylic cement (Acrifix 192 kind), the joint turned and polished: a faint line at the equator (to judge on the first sample); a 40 port cut at its underside. One part where there were three: the frost scatters because it faces air



FRONT



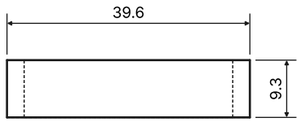
FROM THE RIGHT



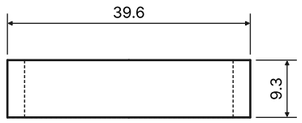
FROM ABOVE

2. grommet: lathe, aluminium; 3 cm3 each; 1 for the set; 1:1

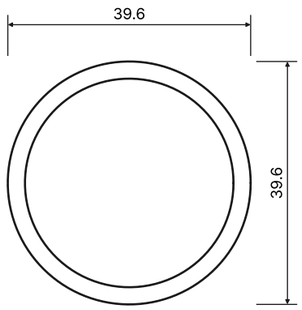
black anodised aluminium: lines the port, bonded



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

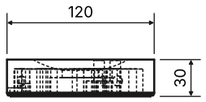
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

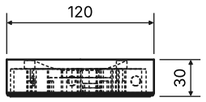
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. base: lathe, cast-acrylic; 160 cm3 each; 1 for the set; 1:5

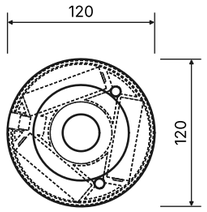
black cast acrylic, polished, turned from rod: 120 × 30 (round 32f's), the cup turned in it; from below the hatch's counterbore (112), the middle pocket (50) for the chassis, the board and the LED, the three cells' bays, the ballast's corners and the cable notch



FRONT



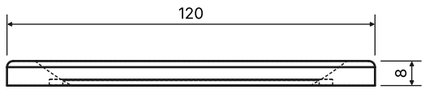
FROM THE RIGHT



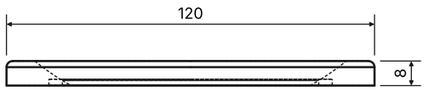
FROM ABOVE

4. ring: lathe, aluminium; 36 cm3 each; 1 for the set; 1:2

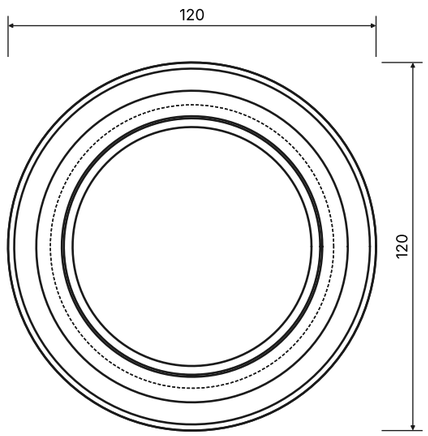
aluminium, black anodised, satin, insulated from the chassis: the touch (tap to switch, hold to dim, 2200 to 2700 K)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

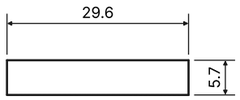
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

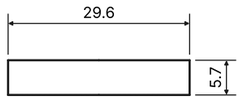
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. window: lathe, cast-acrylic; 4 cm3 each; 1 for the set; 1:1

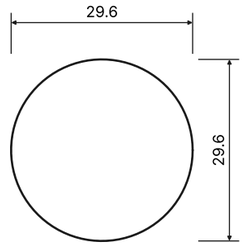
a clear acrylic plug, sealed with an O-ring: the LED under it, the base IP44 with the ball off



FRONT



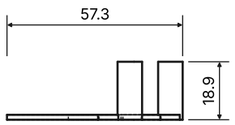
FROM THE RIGHT



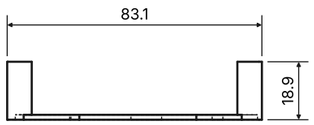
FROM ABOVE

6. chassis: mill, aluminium; 5 cm3 each; 1 for the set; 1:2

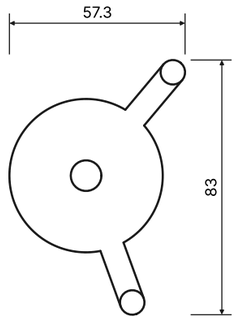
aluminium 6061, 50 across: the LED's heat spreader and the board's carrier, screwed up into the base on threaded inserts, an O-ring between. Review 2026-11-01: two arms 8 × 1.6 out to r 44 in the cell-free corners and a Ø8 post at each up to the touch ring (on a 0.5 insulating thermal pad): about 0.4 W/K to the ring, so about 4 K across the path at 1.5 W; the ring's 9,500 mm2 to the room about 16 K more [E]: the chassis about 20 K over the room at full while charging, to prove on the bench



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

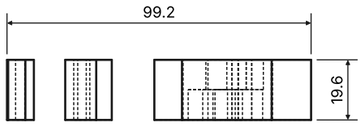
mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

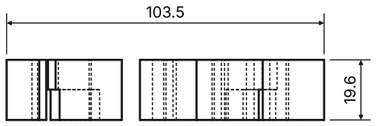
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. ballast: mill, steel; 17 cm3 each; 1 for the set; 1:2

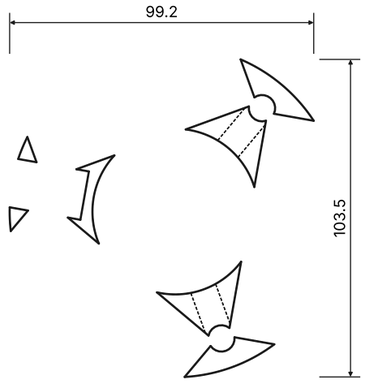
mild steel, zinc-plated: three corner pieces between the cells' bays, screwed up into the base



FRONT



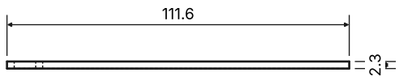
FROM THE RIGHT



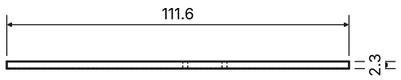
FROM ABOVE

13. hatch: mill, aluminium; 22 cm3 each; 1 for the set; 1:2

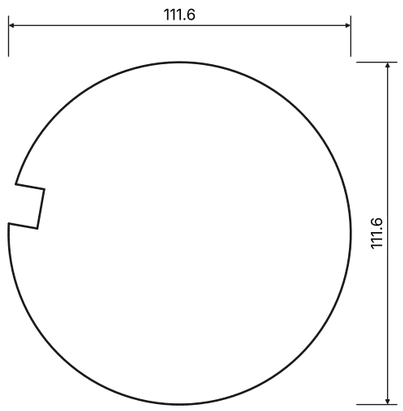
black anodised aluminium, 2.3 thick, in a counterbore under the base: four captive M3 screws (Phillips or Torx: commercially available tools), an O-ring (IP44); a label inside: the cells' type (LiFePO4 18650 only) and how to change them



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	634 cm3, 170 × 170 × 167	1,546.84
2	grommet	lathe	1	Aluminium 6061	50 × 50 × 19	46.19
3	base	lathe	1	Cast acrylic tube, tinted through	130 × 130 × 40	112.04
4	ring	lathe	1	Aluminium 6061	130 × 130 × 18	52.61
5	window	lathe	1	Cast acrylic tube, tinted through	40 × 40 × 16	45.98
6	chassis	mill	1	Aluminium 6061	67 × 93 × 29	49.52
7	pads	buy	1	Bergquist or 3M (to choose) Review 2026-11-01: two silicone gap pads Ø8 × 0.5, about 3 W/m K, electrically insulating (over 5 kV/mm), between the chassis's posts and the touch ring		1.00
8	ballast	mill	1	Mild steel	109 × 114 × 30	48.67
9	board	buy	1	to make (a contract PCB house) Core's board since round 41 (the 30 base), revised by the review of 2026-11-01: a USB PD sink at 9 V on the USB-C path and the dock's pins (an ideal diode on each input); the LTC4162-F LiFePO4 charger with power path		34.00
10	led	buy	1	Nichia or Seoul (to choose) A 30 mm round warm-dim LED board, two channels (2200 K and 2700 K, CRI 90+), 130 lm/W or better: 200 lm at 1.5 W or less (review 2026-11-01; was 2.5 W), on the aluminum chassis's heat path to the touch ring; a		9.00
11	cells	buy	1	Power Sonic PSL-FP-IFR18650EC (round 42), three in parallel (1S3P): LiFePO4 18650 energy cell, 3.2 V, 1.5 Ah rated, 4.8 Wh each, 14.4 Wh the set; 2,000 cycles at 100 % depth of discharge; 65 × 18.2 mm (flat top on its datasheet: Core needs a		9.00
12	holder	buy	1	to make (in the milled base) No holder (review, 2026-11-01): bare spring contacts in the milled base. The − end has a nickel-plated conical spring in a pocket; the + end has a gold-flashed leaf at the bottom of an 8 mm shroud 1.5 deep, so only a		3.00
13	hatch	mill	1	Aluminium 6061	122 × 122 × 12	49.55
14	usb	buy	1	Two EPDM O-rings (the window, the chassis to the drum), a silicone foot ring 110 x 2 (Shore 40, non-slip), a sealed USB-C receptacle (IP67 rated, e.g. GCT USB4135 kind): IP44		5.00
15	seat	buy	1	A silicone seat ring 92 × 84 × 1.5 (Shore 40, black) in the base's cup: the ball rests on it, and it stops a nudge rolling it		1.50
16	foot-ring	buy	1	A silicone foot ring 117 × 114 × 2 (Shore 40, non-slip), and the hatch's O-ring		2.00
17	cable	buy	1	any (made to order) A slim black braided-fabric USB-C to USB-C cable, 2 m, 3 A, a black right-angle overmoulded plug for the notch		6.00
18	adapter	buy	1	any (UL/CE listed) A 20 W USB-C wall adapter, black, UL 62368-1 listed, DOE Level VI external power supply: it is part of Core's battery charger system as tested (the DOE standard and California Title 20, listed in the CEC's MAEDbS under Core's		7.00
19	screws	buy	1	Four captive M3 × 6 stainless pan-head screws (Phillips or Torx T10) with retaining washers, for the hatch		1.50

Parts list

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. Core (the Orb until round 35): a crystal ball, a 160 tinted ball low in the cup of a 120 × 30 base (round 41), 180 tall. The ball lifts off: the LED is under a sealed window in the base, so balls swap colour by hand.
2. One ball where there were three layers: tinted cast acrylic, two halves formed on one mould and joined at the equator, their inside bead-blasted. The frost faces air, so it scatters; seen from outside it is behind the tint, and the port that would show it faces the base.
3. Cordless (the owner, 2026-10-31): three button-top LiFePO4 18650 cells (Power Sonic PSL-FP-IFR18650EC class, 14.4 Wh) on bare spring contacts in bays milled round the LED, behind a hatch underneath on four captive screws, the user's to replace with standard cells. It charges on its dock or from USB-C underneath at 9 V (USB PD, the 20 W adapter in the box; CORE-RECHARGE.md), and charges while it runs, about 2.7 hours to 90 % at 1.5 A. About five hours a charge (the copy until the bench proves more: computed, about 7.8 hours at full brightness, 1.5 W, at the default charge limit of 90 %, counted by a coulomb counter).
4. Long life: LiFePO4 keeps about 80 % of its capacity after about 2,000 cycles (ordinary Li-ion about 500); the charge limit, a battery-management circuit (protection, a coulomb counter, each cell behind back-to-back FETs off until checked, charging held off below 0 C and above 45 C by the cells' NTC, thermal foldback) and no exposed live contacts. To be listed as one product to UL 153 with Supplement SA; the cells to IEC 62133-2 and UN 38.3.
5. Touch the black ring: tap to switch, hold to dim, 2200 to 2700 K as it dims. IP44: the base sealed by O-rings at the window, the chassis and the hatch, the USB-C receptacle sealed; with the ball on or off.
6. Black, so the tint is the only colour: the base black cast acrylic, polished, over an aluminium chassis and a steel ballast; the ring, the port's lining and the hatch black anodised aluminium.

Notes

Core: Table lamp in cast acrylic, a crystal ball, plugged in or on its own cells

mm | issue 2026-11-01 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15